

Gelei Xu

Curriculum Vitae

Department of Computer Science and Engineering
College of Engineering
University of Notre Dame
Notre Dame, IN 46556 USA

Website: <https://gracellgg.github.io/>

Email: gxu4@nd.edu

Research Interests

AI for Healthcare, personalized AI, and on-device AI.

Education

University of Notre Dame

Ph.D. Student in Computer Science and Engineering

Advisor: Yiyu Shi

2023 – Present

Notre Dame, IN, USA

Southern University of Science and Technology

B.E. in Computer Science and Engineering

Advisor: Jiang Liu

2019 – 2023

Shenzhen, Guangdong, China

Selected Honors and Awards

IDEA Center Edison Innovation Fellowship (\$16,500)	2025–2026
MICCAI 2025 NIH Registration Grant (\$700)	2025
Graduate School Professional Development Awards (\$3750)	2024, 2025
DAC Young Fellow	2024
Lyman Taylor & W Stuart Travel Award (\$1500)	2024
Excellent Graduate in the SUSTech (Top 10%)	2023
Enterprise Intelligence Base Scholarship (¥5000)	2022

Major Refereed Conference and Journal Papers

- [1] **Rethinking Fairness in Medical Imaging: Maximizing Group-Specific Performance with Application to Skin Disease Diagnosis**
Gelei Xu, Yuying Duan, Jun Xia, Ching-Hao Chiu, Michael Lemmon, Wei Jin, and Yiyu Shi
Medical Image Analysis (2025, IF 11.8)
- [2] **Tiny-Align: Bridging Automatic Speech Recognition and Large Language Model on Edge**
Ruiyang Qin, Dancheng Liu, Gelei Xu, Amir Nassereldine, Zheyu Yan, Chenhui Xu, Yuting Hu, Shaocong Wang, X. Sharon Hu, Jinjun Xiong, and Yiyu Shi
Proceedings of the 2025 International Conference on Computer-Aided Design (ICCAD 2025, acceptance rate 24.7%)

- [3] **Incorporating Rather Than Eliminating: Achieving Fairness for Skin Disease Diagnosis Through Group-Specific Experts**
Gelei Xu, Yuying Duan, Zheyuan Liu, Xueyang Li, Meng Jiang, Michael Lemmon, Wei Jin, and Yiyu Shi
Proceedings of the 2025 Medical Image Computing and Computer Assisted Interventions (MICCAI 2025, early acceptance, acceptance rate 9%)
- [4] **The Cost of Local and Global Fairness in Federated Learning**
Yuying Duan, Gelei Xu, Yiyu Shi, and Michael Lemmon
Proceedings of the 2025 International Conference on Artificial Intelligence and Statistics (AISTATS 2025)
- [5] **Enabling On-Device Learning via Experience Replay with Efficient Dataset Condensation**
Gelei Xu, Ningzhi Tang, Jun Xia, Ruiyang Qin, Wei Jin, and Yiyu Shi
Proceedings of the 2024 Design, Automation & Test in Europe Conference & Exhibition (DATE 2024, acceptance rate 25%)
- [6] **Automatic Cortical Cataract Classification Framework Based on AS-OCT Images**
Gelei Xu*, Xiaoqing Zhang*, Zunjie Xiao, Risa Higashita, Wan Chen, Jin Yuan, and Jiang Liu
Computer Systems & Applications (CSA), 2022, 31(12).
- [7] **A Novel Local-Global Spatial Attention Network for Cortical Cataract Classification in AS-OCT**
Zunjie Xiao, Xiaoqing Zhang, Qingyang Sun, Zhuofei Wei, Gelei Xu, Yuan Jin, Risa Higashita, and Jiang Liu
Proceedings of the Pattern Recognition and Computer Vision: 5th Chinese Conference (PRCV 2022).
- [8] **Channel-Wise and Spatial Feature Recalibration Network for Nuclear Cataract Classification**
Xiaoqing Zhang, Gelei Xu, Junyong Shen, Zunjie Xiao, Qiuyang Yan, Jin Yuan, Risa Higashita, Jiang Liu
Proceedings of the 2022 IEEE International Conference on Multimedia and Expo (ICME 2022)
- [9] **CCA-Net: Clinical-awareness Attention Network for Nuclear Cataract Classification in AS-OCT**
Xiaoqing Zhang, Zunjie Xiao, Lingxi Hu, Gelei Xu, Risa Higashita, Wan Chen, Jin Yuan, Jiang Liu
Knowledge-Based Systems (KBS), 2022, 250.

Lightly Reviewed Posters, Extended Abstracts, and Workshop Papers

- [1] **Achieving Fairness in Dermatological Disease Diagnosis Through Automatic Weight Adjusting Federated Learning and Personalization**
Gelei Xu, Yawen Wu, Zheng Jia, Jingtong Hu, and Yiyu Shi
Proceedings of the MICCAI 2025 Workshop on Fairness of AI in Medical Imaging (FAIMI)
- [2] **An Adaptive System for Wearable Devices to Detect Stress Using Physiological Signals**
Gelei Xu, Ruiyang Qin, Zhi Zheng, and Yiyu Shi
Proceedings of the 2024 Workshop on Conference on Human Factors in Computing Systems (CHI)

Preprint Papers

- [1] **NaturalEdit: Code Modification through Direct Interaction with Adaptive Natural Language Representation**
Ningzhi Tang, David Meininger, [Gelei Xu](#), Yiyu Shi, Yu Huang, Collin McMillan, Toby Jia-Jun Li
Submitted to Conference on Human Factors in Computing Systems (CHI) ([link](#))
- [2] **AT-CXR: Uncertainty-Aware Agentic Triage for Chest X-rays**
Xueyang Li, Mingze Jiang, [Gelei Xu](#), Jun Xia, Mengzhao Jia, Danny Chen, Yiyu Shi
Arxiv ([link](#))
- [3] **One Training Fits All: Generalized Data Condensation via Mixture-of-Information Bottleneck Guidance**
Jun Xia, Zixuan Pan, [Gelei Xu](#), Yiyu Shi
Arxiv ([link](#))
- [4] **Analysis: Serving Individuals with Language Impairments using Automatic Speech Recognition Models and Large Language Models: Challenges and Opportunities**
Yiyu Shi, Ruiyang Qin, Haoxinran Yu, Lixuan Wei, Yuxuan Liu, Dancheng Liu, Chenhui Xu, Jiajie Li, [Gelei Xu](#), Ahmed Abbasi, Jinjun Xiong, Xiufan Yu, Zhi Zheng
Submitted to Nature Communications ([link](#))
- [5] **A Survey of AI Agents in Healthcare**
[Gelei Xu*](#), Xueyang Li*, Yixiong Chen*, Yuying Duan*, Shuqing Wu*, Alexander Yu*, Ching-Hao Chiu*, Juntong Ni*, Ningzhi Tang, Toby Jia-Jun Li, Alan Yuille, Wei Jin, Yiyu Shi
Techrxiv ([link](#))

Teaching Experience

Graduate Associates for Teaching ([link](#)) 2025 – 2026
Kaneb Center for Teaching Excellence, University of Notre Dame

- **Instructor**, *Foundations of Teaching Workshop Series* ([Part 1](#), [Part 2](#), [Part 3](#), [Part 4](#))
- **Instructor**, *Assessing Prior Knowledge Workshop* ([link](#))

Guest Lecturer, *EE 60572 Introduction to Deep Learning* ([link](#)) April 24, 2025
Department of Electrical Engineering, University of Notre Dame (host: Michael Lemmon)

Teaching Assistant, *CSE 20221 Logic Design* Spring 2024
Department of Computer Science and Engineering, University of Notre Dame

Teaching Assistant, *CSE 40535 Computer Vision* Fall 2023
Department of Computer Science and Engineering, University of Notre Dame

Service

Committee Member, College of Engineering Grad Student Leadership Committee, 2024 – 2025
Graduate Recruitment Ambassador, Notre Dame College of Engineering, NDiSTEM 2025
Invited Panelist, Undergraduate Outreach Program, Notre Dame College of Engineering, 2025
Mentor Judge, NDiSTEM 2025 Research Presentations
Reviewer, TOIS, TECS, TCPS, MICCAI, RO-MAN, FSS